IT University of Copenhagen

0.0.0.1 Software Architecture Reconstruction

1 III: Evolutionary Analysis for Architecture Recovery

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github.com/mircealungu/reconstruction

Figure 1: 400

> No man ever steps in the same river twice, for it is not the same river and he is not the same man.
>
> – Heraclitus

2 Why *Must* Software Systems Evolve?

> An e-type program that is used in a real-world environment must change, or become progressively less useful in that environment.
>
> – M. Lehman, *The Law of Continuing Change*

2.1 E-Type Systems

In the terminology of Lehman, the "e" in *e-type* stands for *embedded* in the real world. Since the world changes, the system must change too.

There are actually two parts of the "world" where changes can impact a given software system. The human context of use and the technical context.

2.1.1 The Human Context

To think about: *Do you have good examples of systems that had to change because the real world changed around them?*

2.1.1.0.1 Examples?

- the software that computes taxes in Denmark
- all the websites in Europe once GDPR came into being
- etc.

2.1.2 The Technical Context

Think about the `npm` ecosystem. Every day you can execute `npm audit` in your React-based web application to find out that a dozen of the packages you depend on have new versions. Do you upgrade? Do you stay like this for a while? What's the best strategy for managing this portfolio of dependencies?

But this is not only about the `npm` ecosystem. The same situations emerges when a developer builds a system on top of other libraries, and even programming languages change. If one is not keeping up with the evolution of their programming language, their code will sooner or later stop working. How could this be? Some of the libraries their code depends on, might have dropped support for that version of the language. So they either upgrade, or they are left behind. If a developer des not want to be left behind, they have to keep up with all their upstream dependencies.

Developing software is a little bit like living in Wonderland where the Red Queen is telling Alice:

> "*In this place you have to run to stay in one place*".

2.2 Non-e-type systems

To think about: *What other kinds of systems are there then? Can you think about another type? Are there programs that are not impacted by the change in the world around them?*

2.2.0.1 Examples?

- a chess engine

- a red-black tree balancing algorithm
- ...

It seems that one could even argue that this is the difference between algorithms and software systems: algorithms don't have to change with the world, while software systems have to.

3 Software Evolution

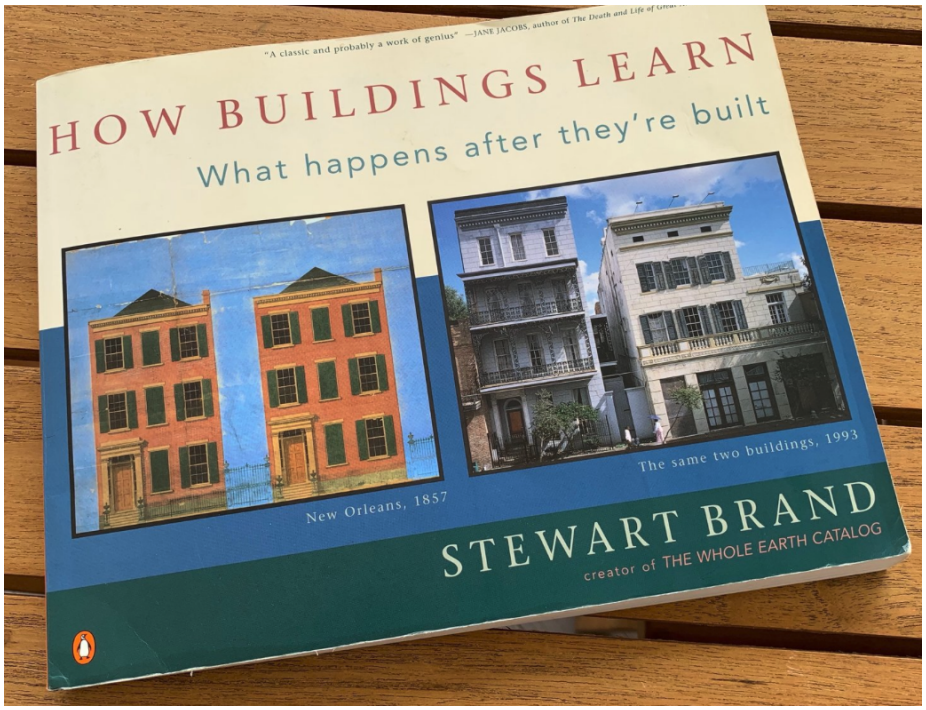
Software evolution is the continual development of a piece of software after its initial release to address changing stakeholder requirements.

Observations: - It used to be called *software maintenance* - Nowadays evolution is the preferred term because it highlights the fact that a software system is never *finished*

4 Does Architecture Evolve?

From this POV, the **architecture metaphor might not be the best** - because we normally think about architecture as unchanging.

This is where Stewart Brand – one of my favorite non-fiction authors – comes to help. He wrote a whole book about how buildings evolve over time exactly with the goal of making it clear that architecture is also changing.



I personally think a ***garden*** **would have been a better metaphor** for a software system. You must constantly tend to your garden if you want to maintain it.

5 Version Repositories Can Help Re-trace Evolution

One of the benefits of the widespread adoption of tools like git is the fact that the meta-data captured in the version control system captures relevant information about the evolution of the system.

There are **multiple kinds of information that might be relevant for the architecture reconstruction** process that can be recovered from the version control system (VCS). Several are presented in this section.

5.1 Logical Coupling: the parts of the system that always change together

When two entities *frequently* change together, even if there is no explicit dependency between them, we call that between them there is **logical coupling**.

This information can be inferred from the version control. The concept was introduced in 1998 in a paper by Gall et al. has become quite popular in the

meantime. Adam Tornhill has a tool that computes it and wrote a book about many of the concepts discussed in this course. Other tools that compute it exist.

However, defining the concept is a challenge, because it's after all a matter of selecting thresholds and constants: - how many changes should two entities have together before we call them coupled? - what percentage of changes can be not *together* while still allowing us to consider them coupled?

Pros

- language-independent method
- can even detect dependencies between parts of the code that are not written in the same language
- can detect dependencies that are indirect

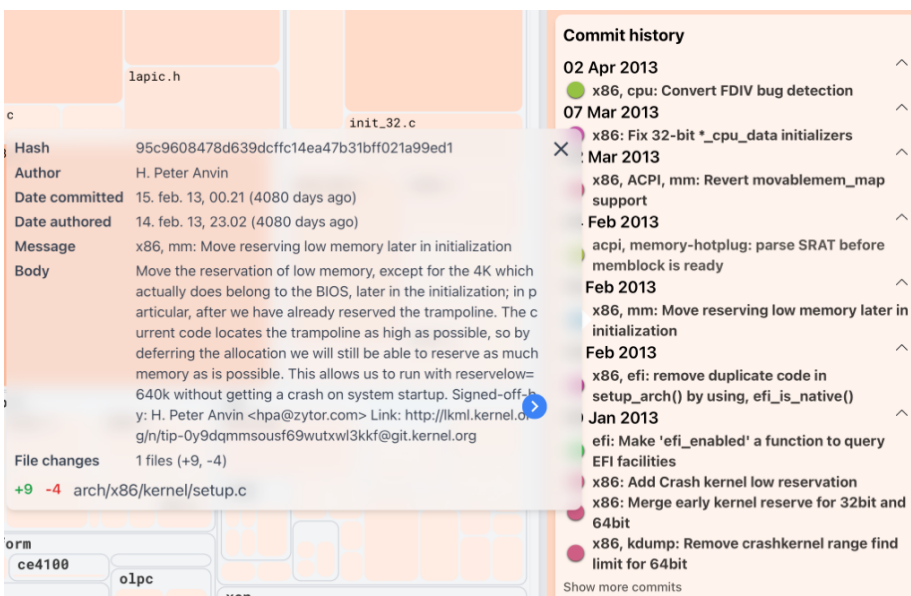
Cons

- only a small part of the dependencies can be detected this way

5.2 Documentation: fine-grained, based on `git` comments

One of the beautiful insights I recently had is the importance of the git messages associated to commits as documentation.

Indeed, even if there is no separate documentation, well described commits can serve as an evolving documentation for a software system. Look at the following commit comment, from linux, which documents a (+9, -4) change. How many of us are able to write such detailed changes?



However, the information in the git log could be useful for architecture recovery, only if the developers wrote meaningful messages.

5.3 Churn: highlighting the parts of the system are most changed over time

Churn is a metric that indicates how often a given piece of code gets edited.

- process metric (*as opposed to? do you remember the alternate concept?*)
- can be detected with **language independent analysis** (which is good for polyglot systems)
- can be applied to all kinds of code —e.g., a file, a class, a function—

Why would places in the system with **high-code churn** be relevant? - are likely to be most important parts of the code - studies observe correlation between *code churn* and complexity metrics - high *code churn* predicts bugs better than size - it's likely that they'll require more effort in the future (e.g. yesterday's weather [Girba et al.])

Evolutionary Hotspots – **an architectural viewpoint that highlights those code entities with high churn**.

Notebook: Abstracting Churn Along the Module Hierarchy in Python

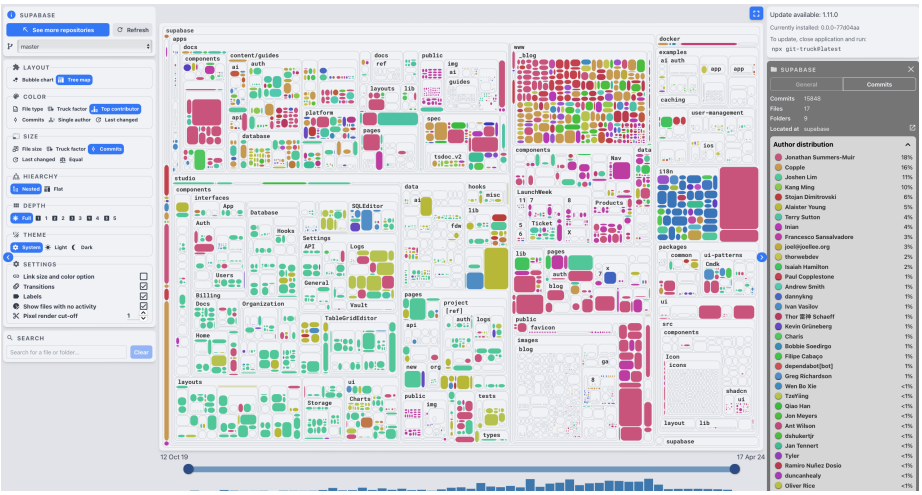
Keep in mind when using churn in architecture recovery

1. Caveat: **one must take into account developer styles**, e.g. the micro-commits developer vs. the large chunk committer
   - you could use LOC instead of # number of commits;
   - what could the problems with this be?
2. Removing irrelevant files that change frequently (`README.md`, or `LICENSE.md`)
   - Combine with static complexity metrics
   - Manual investigation
3. Selecting the appropriate time-interval for the analysis
   - Weighting towards recency (discarding past changes more)
4. Tracking file renames over the course of a system's history
   - Sometimes git loses track of file history: e.g. if you rename and make changes at the same time

6 Bonus: Estimating Code Ownership Based on Git Repositories

This can help find an expert for a given problem and might also provide an estimate of the maintainability related risks of a given project.

Compare the following two visualizations. Which of the two projects would you consider to be more maintainable, everything else being equal?



The images are generated with Git-Truck – a git repository visualization tool developed here at ITU. To try it out on your own, use:

```
git clone git@github.com:zeeguu/api.git
npx -y git-truck-beta@latest
```

Why is the tool defined this way? Because of the **truck factor**: "*the number of people on your team that have to be hit by a truck (or quit) before the project is in serious trouble*". *A Novel Approach for Estimating Truck Factors*, by Avelino et al. discusses some of the challenges of formally defining the concept.

7 For Your Projects

Consider adding an evolutionary analysis component to your project. - churn metrics - code-ownership analysis

8 To Think About

https://gitdiagram.com/ – stupid idea. But popular. But still stupid. Could you make it smarter with the knowledge in this course?

What if you could replay the history of a system from the beginning but only showing those files that made it to the end. So project the beginnings through the perspective of the endings. Would that be a useful way of focusing on the most relevant aspects of the system?

9 References

Detection of Logical Coupling Based on Product Release History, by Harald Gall, Karin Hajek, and Mehdi Jazayeri.

Laws of Software Evolution Revisited. M. M. Lehman